Olbrecht 2021 HGSTOC — analysis summary

RUN 2 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(sample) swept over θ×n_pcs with per-neighbourhood iLISI; at θ=10 iLISI plateaus at 3.34/6 and the 6 samples form one intermixed, connected manifold (single component; 0 islands removed).

Gate 2 — module geography → GO (non-branching)

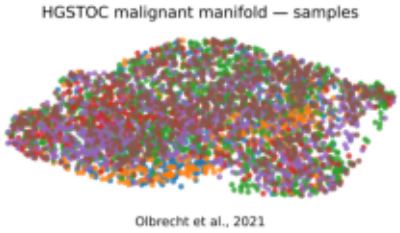
Gast & pSom co-vary (smoothed r=+0.44) at the root pole; Crest is an inverse terminal gradient (Crest~pSom r=−0.68, Crest-high decile mean pseudotime 0.87). A single embryonic axis, not two competing arms.

Gate 3 — cross-repression → NA

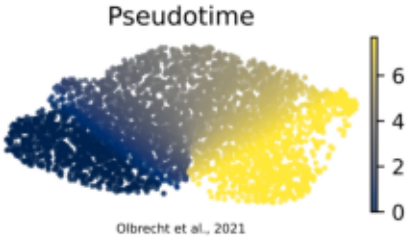
Branching-only. No bifurcation (PPT forks scale with node count 6→13 = noise; curve fits 0 forks, fold-free ratio 2.70), so there is no hub to test for Gast↔pSom mutual exclusivity.

STEP FIGURES

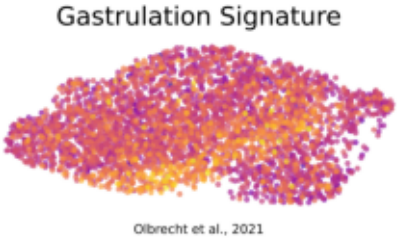
Manifold — samples



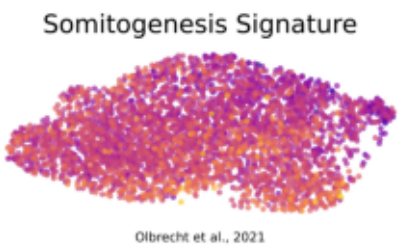
Pseudotime



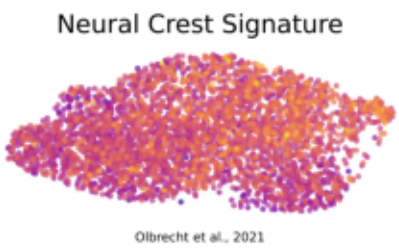
Gastrulation



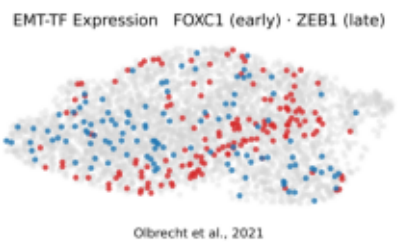
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

High-grade serous tubo-ovarian carcinoma (Olbrecht et al., 2021), 10x scRNA-seq — 3,936 malignant cells across 6/10 samples (≥155-cell threshold) enter the analysis; the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.

The embryonic pattern reproduces as a single collapsed axis: a Gast/pSom co-expressing developmental origin resolving into a neural-crest-high terminus (Gate 1 GO / Gate 2 GO / Gate 3 NA). Pseudotime correlations Gast ρ=−0.41, pSom −0.18, Crest +0.47 (smoothed). Linear-pSom/Crest geometry; Gast & pSom co-vary rather than diverge, so there is no Gast–pSom bifurcation (Gate 3 NA). EMT-TFs FOXC1 (root) → ZEB1 (terminus).

Parameter choices

Min-cell threshold	155 (geo-mean of 384→63 count gap, ratio 6.1) → 6/10 samples	**Integration**	Harmony(sample), θ=10, n_pcs=20 (iLISI plateau 3.34/6)
Island removal	radius graph 6× median NN → 0 cells removed (single component)	**Geometry**	Linear-pSom/Crest — curve; 2 tips, 0 forks, fold-free 2.70
Root	Gast/pSom-high tip (developmental origin; Crest is late)	**Pseudotime ρ**	Gast −0.41, pSom −0.18, Crest +0.47 (smoothed)
Overdispersed genes	1,413 (scFates find_overdispersed)	**EMT-TFs**	FOXC1 (early/root) → ZEB1 (late/terminus)